

	$= 1.5 \times 10^5 \text{ V m}^{-1}$
4(b)(ii)	Magnetic force is directly proportional to velocity but electric force does not depend on velocity. For a proton, the magnetic force on the charge at it's point of entry is upwards. Hence when the proton's velocity is increase, it will be deviated upwards.
5	At zero kelvin, the conduction band does not have mobile electrons and the valence band is fully occupied. When an external electric field is applied, there are no mobile charge carriers for conduction. Hence the resistance is high. The energy band gap of an intrinsic semiconductor is small (only around 1 eV). When temperature increases, electrons in the valance band are thermally excited and move to the conduction band, leaving behind holes in the valence band.

	When an external electric field is applied, mobile electrons in the conduction band move in the direction opposite to the electric field, while holes (which behaves like positive charges) in the valence band move in the same direction of the electric field, and a current is formed. As the temperature increases, more electron-hole pairs are formed and the conductivity of the semiconductor increases. Hence resistance decreases.
6(a)(i)	If the daughter is radiative, it will decay and the count rate measured will not be due to the decay of the parent. Q_2 also